

Eric Elmoznino

Artificial Intelligence
Cognitive Neuroscience

Contact

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Websites



Education

PhD | Computer Science
Mila (Université de Montréal)
2022-Now GPA: 4.3/4.3

MA | Cognitive Science
Johns Hopkins University
2019-2020 GPA: A+

BASc | Computer Engineering
University of Toronto
2014-2019 GPA: 3.95/4.0

DCS | Health Sciences
Dawson College
2012-2014 GPA: 96/100

Skills

Languages/Frameworks

Python, PyTorch, JAX, TensorFlow,
sklearn, Matlab, C++, C, C#, Swift,
JavaScript, HTML, CSS, LaTeX

Subjects/Techniques

AI, Deep learning, Computational
neuroscience, Linguistics,
Kolmogorov complexity

Spoken Languages

English, French (Fluent in both)

Interests

Mind, Brain, and AI

Consciousness, Compositionality,
In-context learning, Causality,
Knowledge acquisition, OoD
generalization, Lifelong learning

Public Speaking

Technical presentations, Teaching

Other Disciplines

Data Science, Genetics, Astrophysics,
Philosophy of mind, Epistemology

Hobbies

Reading, Piano, Meditation,
Snowboarding, Tennis, Basketball

Research Positions & Work Experience

- 2022-Now **AI Researcher (PhD)** Mila (Université de Montréal), Montreal, QC
Development of AI with inductive biases from conscious higher-level cognition, supervised by Profs. Guillaume Lajoie and Yoshua Bengio
- 2024-Now **AI Student Researcher** Google – Paradigms of Intelligence, Remote
Unifying action and prediction in large generative sequence models, supervised by João Sacramento
- 2025 **Visiting Researcher** University of Suwon, South Korea
Systems and Intervention Neuroscience group, supervised by Prof. Hakwan Lau ◦ Project on brain-computer interfaces using fMRI and generative models ◦ Project on theories of consciousness
- 2020-2025 **Data Science Instructor** Lighthouse Labs, Remote
Teach lectures on machine learning topics at a full-time data science bootcamp to students with no prior coding experience
- 2019-22 **Cognitive Science Researcher (MA)** Johns Hopkins University, Baltimore, MD
Research on information representation and algorithms in the visual system of the human brain with Prof. Michael Bonner
- 2021 **Computational Neuroscience TA** Neuromatch Academy, Remote
Lead groups of students through tutorial exercises ◦ Review lecture material and answer student questions
- 2017-19 **Machine Learning Researcher** ModiFace, Toronto, ON
Work on computer vision machine learning models for the beauty industry ◦ Research papers on makeup rendering and skin condition diagnostics using deep learning
- 2018 **Computer Vision Contractor** Precious, Remote
Work on computer vision machine learning models related to facial perception for a mobile app that automatically makes photo albums of babies for new parents
- 2016 **Software Developer Intern** Orbis Investments, Vancouver, BC
Full-stack web development using AngularJS, Angular Material, ASP.NET MVC, Web API, and SQL Server in order to improve internal workflow efficiency for financial reporting
- 2012-14 **Corporal, Infantry Division** Canadian Armed Forces (Reserves), Valcartier, QC
Participation in combat and reconnaissance exercises

Highlighted work

- 2023-Now **Representational complexity and compositionality**
Deriving a formal quantitative definition of compositionality using ideas from algorithmic information theory
- 2024-Now **Open-ended knowledge growth**
Training a policy with an intrinsic reward inspired by algorithmic information theory to explore and seek new knowledge efficiently
- 2023-Now **Compositional attractor models of human thought**
Learning discrete and compositional models of conscious human thought using neural network attractor dynamics
- 2024-2025 **In-context learning and Occam's Razor**
A normative theory of in-context learning as a meta-learning algorithm for fitting the simplest model that explains the training data
- 2022-23 **Sampling compositions of modular neural networks**
Jointly learning a set of neural network modules and how to sample context-conditioned compositions of them using GFlowNets
- 2021-22 **Dimensionality and Manifold Geometry of Visual Representations**
Quantifying the relationship between the geometry of neural network representations and their similarities to visual cortex

Publications

- 2025 **A Complexity-Based Theory of Compositionality.** [Eric Elmoznino](#), Thomas Jiralerspong, Yoshua Bengio, Guillaume Lajoie *ICML poster*
- 2025 **In-context learning and Occam's razor.** [Eric Elmoznino](#), Tom Marty, Tejas Kasetty, Leo Gagnon, Sarthak Mittal, Mahan Fathi, Dhanya Sridhar, Guillaume Lajoie *ICML poster*
- 2025 **Does learning the right latent variables necessarily improve in-context learning?** [Eric Elmoznino](#), Sarthak Mittal, Leo Gagnon, Sangnie Bhardwaj, Tom Marty, Dhanya Sridhar, Guillaume Lajoie *ICML poster*
- 2025 **Illusions of AI consciousness.** Yoshua Bengio & [Eric Elmoznino](#) *Science*
- 2025 **Multi-agent cooperation through learning-aware policy gradients.** Alexander Meulemans, Seijin Kobayashi, Johannes von Oswald, Nino Scherrer, [Eric Elmoznino](#), Blake Richards, Guillaume Lajoie, Blaise Aguera y Arcas, João Sacramento *ICLR poster*
- 2024 **Amortizing intractable inference in large language models.** Edward J. Hu, Moksh Jain, [Eric Elmoznino](#), Younesse Kaddar, Guillaume Lajoie, Yoshua Bengio, Nikolay Malkin *ICLR talk – best paper honorable mention*
- 2024 **Sources of Richness and Ineffability for Phenomenally Conscious States.** [Eric Elmoznino](#), Xu Ji, George Deane, Axel Constant, Guillaume Dumas, Guillaume Lajoie, Jonathan Simon, Yoshua Bengio *Neuroscience of Consciousness*
- 2024 **High-performing neural network models of visual cortex benefit from high latent dimensionality.** [Eric Elmoznino](#) & Michael F. Bonner *PLOS Computational Biology*
- 2024 **Convolutional architectures are cortex-aligned de novo** Atlas Kazemian, [Eric Elmoznino](#), Michael F. Bonner *Preprint*
- 2023 **Discrete, compositional, and symbolic representations through attractor dynamics.** Andrew Nam, [Eric Elmoznino](#), Nikolay Malkin, Chen Sun, Yoshua Bengio, Guillaume Lajoie *NeurIPS Workshop talk*
- 2023 **Consciousness in Artificial Intelligence: Insights from the Science of Consciousness.** Patrick Butlin, Robert Long, [Eric Elmoznino](#), Yoshua Bengio, Jonathan Birch, Axel Constant, George Deane, Stephen M. Fleming, Chris Frith, Xu Ji, Ryota Kanai, Colin Klein, Grace Lindsay, Matthias Michel, Liad Mudrik, Megan A. K. Peters, Eric Schwitzgebel, Jonathan Simon, Rufin VanRullen *Preprint*
- 2023 **Scene context is predictive of unconstrained object similarity judgments.** Caterina Magri, [Eric Elmoznino](#), Michael F. Bonner *Cognition*
- 2023 **Learning Macro Variables with Auto-encoders.** Maitreyi Swaroop, [Eric Elmoznino](#), Dhanya Sridhar *Neurips Workshop poster*
- 2020 **Visual representations derived from multiplicative interactions.** [Eric Elmoznino](#) & Michael F. Bonner *NeurIPS Workshop poster*
- 2019 **A new procedure, free from human assessment that automatically grades some facial skin structural signs.** Jiang R., Kezele I., Levinshstein A., Flament F., Zhang J., [Elmoznino E.](#), Ma J., Ma J., Coquide J., Arcin V., Omoyuri E., Aarabi P. *International Journal of Cosmetic Science*

Invited Talks

- 2025 Good visual representations are high-dimensional — *Martin Hebart Lab*
- 2024 **Why can't we describe our conscious experiences? An information theoretic attractor dynamics perspective of ineffability** — *Models of Consciousness conference*
- 2024 Consciousness, ineffability, and AI safety — *Mila AI Safety Reading Group*
- 2023 Sampling discrete objects through continuous attractor dynamics — *Mila GFlowNet Reading Group*
- 2023 **Why can't we describe our conscious experiences? An information theoretic attractor dynamics perspective of ineffability** — *Computational Phenomenology Group*
- 2023 **Why can't we describe our conscious experiences? An attractor dynamics perspective of the ineffability of qualia** — *University of Toronto guest lecture*
- 2020 How does the brain work? Cognitive science research — *SABES*
- 2020 Introduction to Programming with Python — *UofTHacks*

Podcasts

- 2025 Could AIs be Conscious? — *New Horizons* podcast
 2023 Why can't we describe our conscious experiences? An information theoretic attractor dynamics perspective of ineffability — *Active Inference Institute* podcast

Supervision

- 2023 **Maitreyi Swaroop** Mila, Montreal, QC
 Masters (Mathematics and Computing)
 2021-22 **Atlas Kazemian** Johns Hopkins University, Baltimore, MD
 Masters (Cognitive Science)
 2020-21 **Adyant Balaji** Johns Hopkins University, Baltimore, MD
 Undergraduate (Computer Engineering & Cognitive Science)
 2019-20 **Maro Maged Doce** Johns Hopkins University, Baltimore, MD
 Undergraduate (Neuroscience)

Patents

- 2022 **System and method for image processing using deep neural networks.** Levinshtein A., Chang C., Phung E., Kezele I., Guo W., Elmoznino E., Jiang R., Aarabi P. *U.S. Patent No. 11216988*. Washington, DC: U.S. Patent and Trademark Office
 2021 **Image-to-image translation using unpaired data for supervised learning.** Elmoznino E., Kezele I., Aarabi P. *U.S. Patent Application No. 17096774*. Washington, DC: U.S. Patent and Trademark Office
 2020 **System and method for augmented reality using conditional cycle-consistent generative image-to-image translation models.** Elmoznino E., Ma H., Kezele I., Phung E., Levinshtein A., Aarabi P. *U.S. Patent Application No. 16683398*. Washington, DC: U.S. Patent and Trademark Office
 2020 **Machine image colour extraction and machine image construction using an extracted colour.** Elmoznino E., Aarabi P., Zhang Y. *U.S. Patent Application No. 16854975*. Washington, DC: U.S. Patent and Trademark Office
 2020 **Automatic image-based skin diagnostics using deep learning.** Jiang R., Ma J., Ma H., Elmoznino E., Kezele I., Levinshtein A., Charbit J., Despois J., Perrot M., Antoinin F., Flament R.S., Parham A. *U.S. Patent Application No. 16702895*. Washington, DC: U.S. Patent and Trademark Office

Scholarships & Awards

- 2024 UNIQUE Conference Travel Award (\$1,000 value)
 2023 Vanier Canadian Graduate Scholarship (\$150,000 value)
 2023 Natural Sciences and Engineering Research Council of Canada (NSERC) PhD Grant (\$105,000 value)
 2022 UNIQUE Neuro-AI Excellence Scholarship (\$15,000 value)
 2016 Class of 4T3 Engineering James Ham Award (\$10,000 value)
 2015 Class of 5T6 Award of Merit (\$15,000 value)
 2013 First Choice Science Award
 2012 McGill Science Award and Scholarship
 2012 A.J. Grant Shield and Scholarship
 2012 Quebec English Public Speaking (Provincial Finals) — Bronze Medal
 2012 Governor General of Canada Academic Medal
 2012 Royal Bank of Canada Shield
 2012 Davies Family Shield
 2012 Eakeley Shield
 2011 Quebec French Public Speaking (Provincial Finals) — Silver Medal

Other Activities

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| 2018 | Instructor for ECE1780 | University of Toronto, Toronto, ON |
| | Taught lectures for a graduate course on DNNs deployed to mobile devices under Prof. Parham Aarabi | |
| 2015-16 | Finance Chair | Electrical and Computer Engineering Club, Toronto, ON |
| | Elected by peers at the University of Toronto to manage the club budget and plan social activities | |
| 2014-15 | Class Representative | Electrical and Computer Engineering Club, Toronto, ON |
| | Elected by peers at the University of Toronto to represent student interest at faculty meetings | |